Customer Shopping Behavior Analysis

This project analyzes 3,900 customer purchases to uncover spending patterns, product preferences, and subscription behavior, guiding strategic business decisions.

Project Overview & Dataset Summary

Project Goal

Analyze transactional data to understand customer behavior, product preferences, and subscription impact for informed business strategies.

Dataset Details

- 3,900 Rows, 18 Columns
- Key Features: Demographics, Purchase Details, Shopping Behavior
- Missing Data: 37 values in Review Rating

Exploratory Data Analysis (Python)

01

Data Loading & Exploration

Imported dataset with pandas; used `df.info()` and `.describe()` for initial insights.

02

Missing Data Handling

Imputed missing Review Rating values using median per product category.

03

Column Standardization

Renamed columns to snake case for improved readability.

04

Feature Engineering

Created `age_group` and `purchase_frequency_days` columns.

05

Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL analysis.

Key Data Insights (SQL)

1

Revenue by Gender

Male customers generated significantly higher revenue ($157,890) than Female customers ($75,191).

2

High-Spending Discount Users

Identified customers who used discounts but still spent above average.

3

Top 5 Products by Rating

Gloves, Sandals, Boots, Hat, and Skirt had the highest average review ratings.

4

Shipping Type Comparison

Express shipping had a slightly higher average purchase amount ($60.48) than Standard ($58.46).

Subscription & Discount Insights

Subscribers vs. Non-Subscribers

Non-subscribers represent a larger group of repeat buyers (2518) compared to subscribers (958).

Discount-Dependent Products

Hat (50%), Sneakers (49.66%), Coat (49.07%), Sweater (48.17%), and Pants (47.37%) were most frequently purchased with discounts.

Customer Segmentation

Loyal Customers

3116 customers, indicating a strong base.

Returning Customers

701 customers, showing potential for increased loyalty.

New Customers

83 customers, a smaller segment for growth.

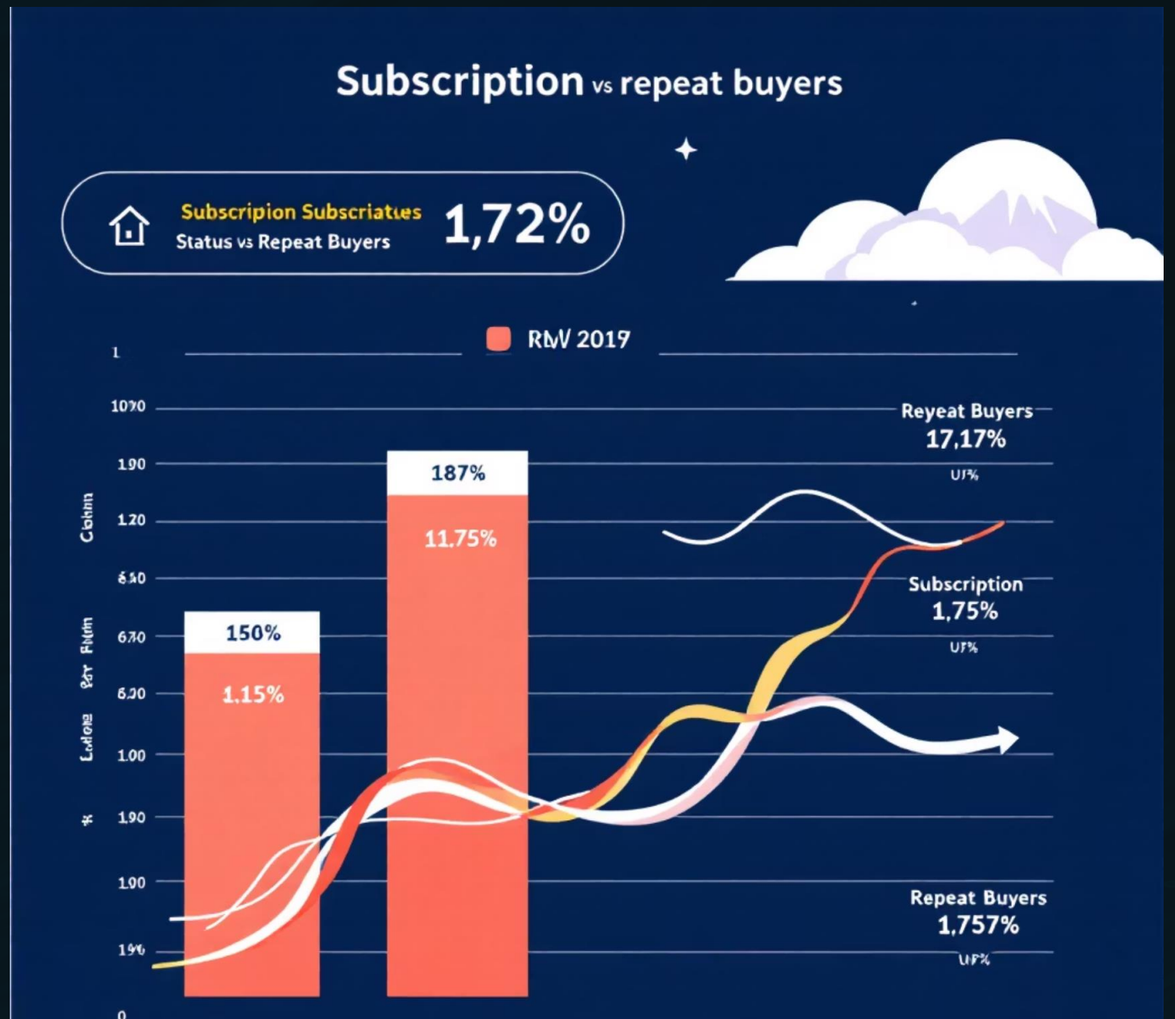
Top Products by Category

1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

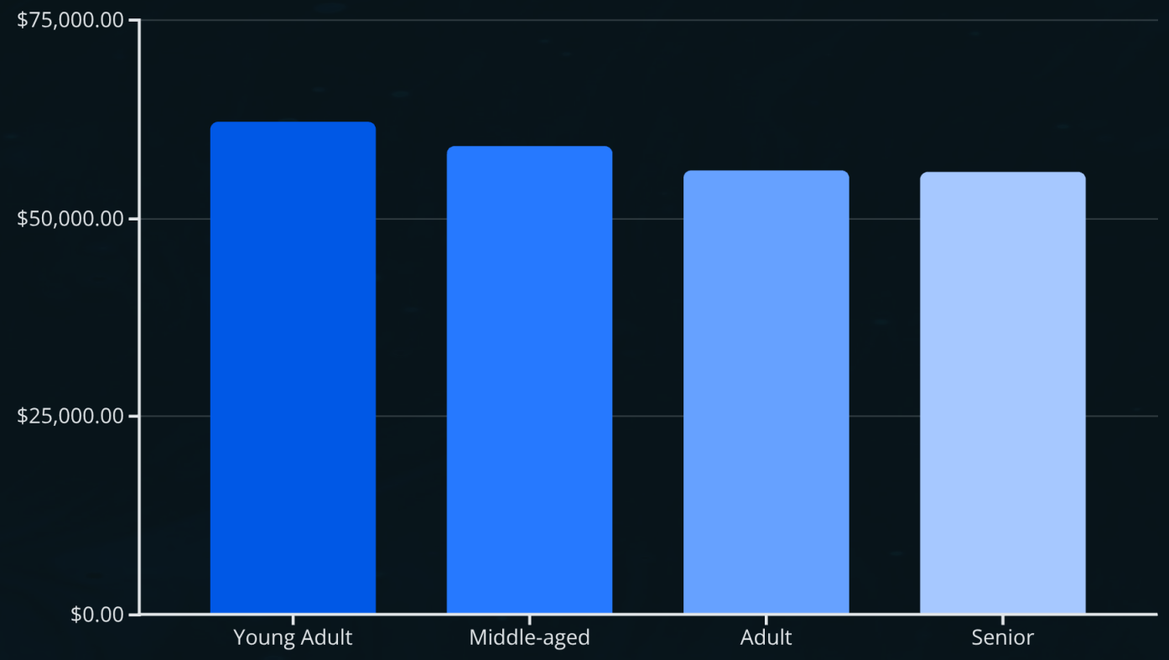
Subscription & Age Group Revenue

Repeat Buyers & Subscriptions

Customers with more than 5 purchases are more likely to be non-subscribers (2518) than subscribers (958).

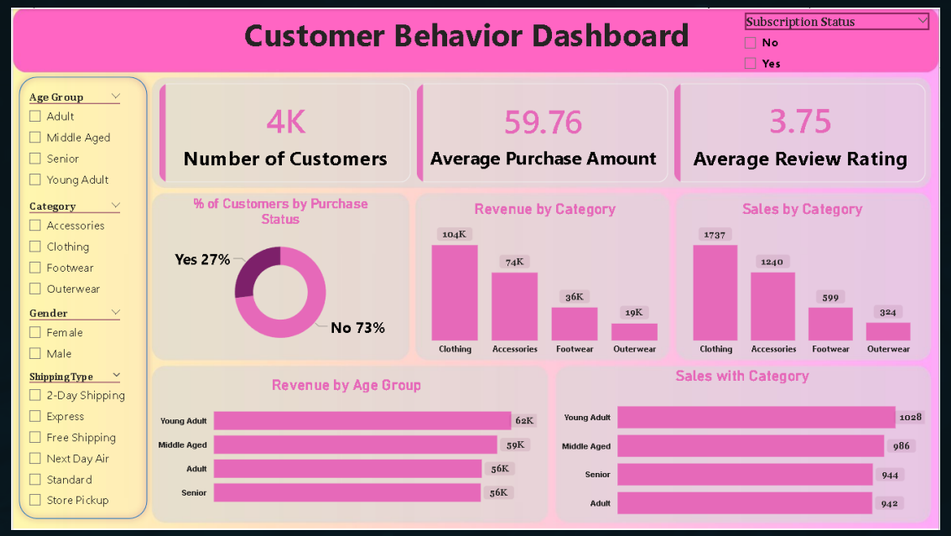


Revenue by Age Group



Power BI Dashboard Overview

The interactive dashboard provides a comprehensive view of customer behavior, allowing filtering by Subscription Status, Age Group, Category, Gender, and Shipping Type.



Key metrics include 4K customers, $59.76 average purchase, and 3.75 average review rating.

Business Recommendations

Target High-Value Age Groups

Focus marketing on Young Adults and Middle-Aged customers with personalized offers.

Reduce Non-Purchase Rate

Introduce first-purchase discounts and retargeting to convert browsing customers.

Optimize Low-Performing Categories

Increase visibility and offer bundles for Footwear and Outerwear.

Improve Customer Satisfaction

Enhance service quality (delivery, packaging, support) to boost repeat purchases.